In [66]:
```
# Format for report:
# (Note: These are example numbers, not the expected values for the input
#
# --- SENSOR REPORT: SENSOR NAME ---
# Valid readings      :    89.3%
# Average reading     :     5.01
# Removed readings    :        3

# Input for demonstration
sensor_name = "sensor a"
sensor_readings = [2.673, -3.876, 4.589, None, None, None, -6.111, 0.435,
                   2.444, 10.897, 0.019, 10.222, -0.543, None, 8.097, 0.9
                   None, 0.765, 0.765, 0.765, 0.765, 7.937, -1.007, 4.405
sensor_threshold = 0.010
```

In [67]:
```
# Cleaning data by removing None and values less then threshold

def clean_sensor_reading (sensor_readings: list [float | None],sensor_thr
    """
    Remove invalid sensor readings.

    A reading is valid if it is not None and is greater
    than or equal to the threshold.

    Args:
        sensor_readings: List of sensor readings.
        sensor_threshold: The minimum value for a valid reading.

    Returns:
        A list with values that are valid readings.
    """
# Making a list to add values that are valid readings.
    cleaned = []

    for reading in sensor_readings:
        if reading is not None and reading >= sensor_threshold:
            cleaned.append(reading)
    return cleaned


# Checking results to see if list is cleaned
result_clean = clean_sensor_reading(sensor_readings, sensor_threshold)
display(result_clean)
print(len(result_clean))
```

```
[2.673,
 4.589,
 0.435,
 2.444,
 10.897,
 0.019,
 10.222,
 8.097,
 0.962,
 1.005,
 0.765,
 0.765,
 0.765,
 0.765,
 7.937,
 4.405]
16
```

In [68]:
```python
#Creating a report to read of information about sensor readings
def sensor_report (sensor_name: str, sensor_readings: [float| None], resu
    """
    Creating a report to read of information about sensor readings

    Args:
        sensor_name: The name of the sensor.
        sensor_readings: The real/orignal list of sensors.
        result_clean: The list of sensors after cleaned and gives valid s

    Returns:
        A string containing that sensor report.
    """
    # calculating valid report results
    valid_reading_count = len(result_clean)
    avg_reading = sum(result_clean)/ valid_reading_count
    valid_reading_percent = valid_reading_count / len(sensor_readings)*10
    removed_reading = len(sensor_readings) - valid_reading_count

    # Creating visual report
    report = f"""--- SENSOR REPORT: {sensor_name} ---
Valid readings      : {valid_reading_percent:.1f}%
Average reading     : {avg_reading:.2f}
Removed readings    : {removed_reading}
"""

# Print the report
print(report)
```

```
--- SENSOR REPORT: sensor a ---
Valid readings      : 53.3%
Average reading     : 3.55
Removed readings    : 14
```